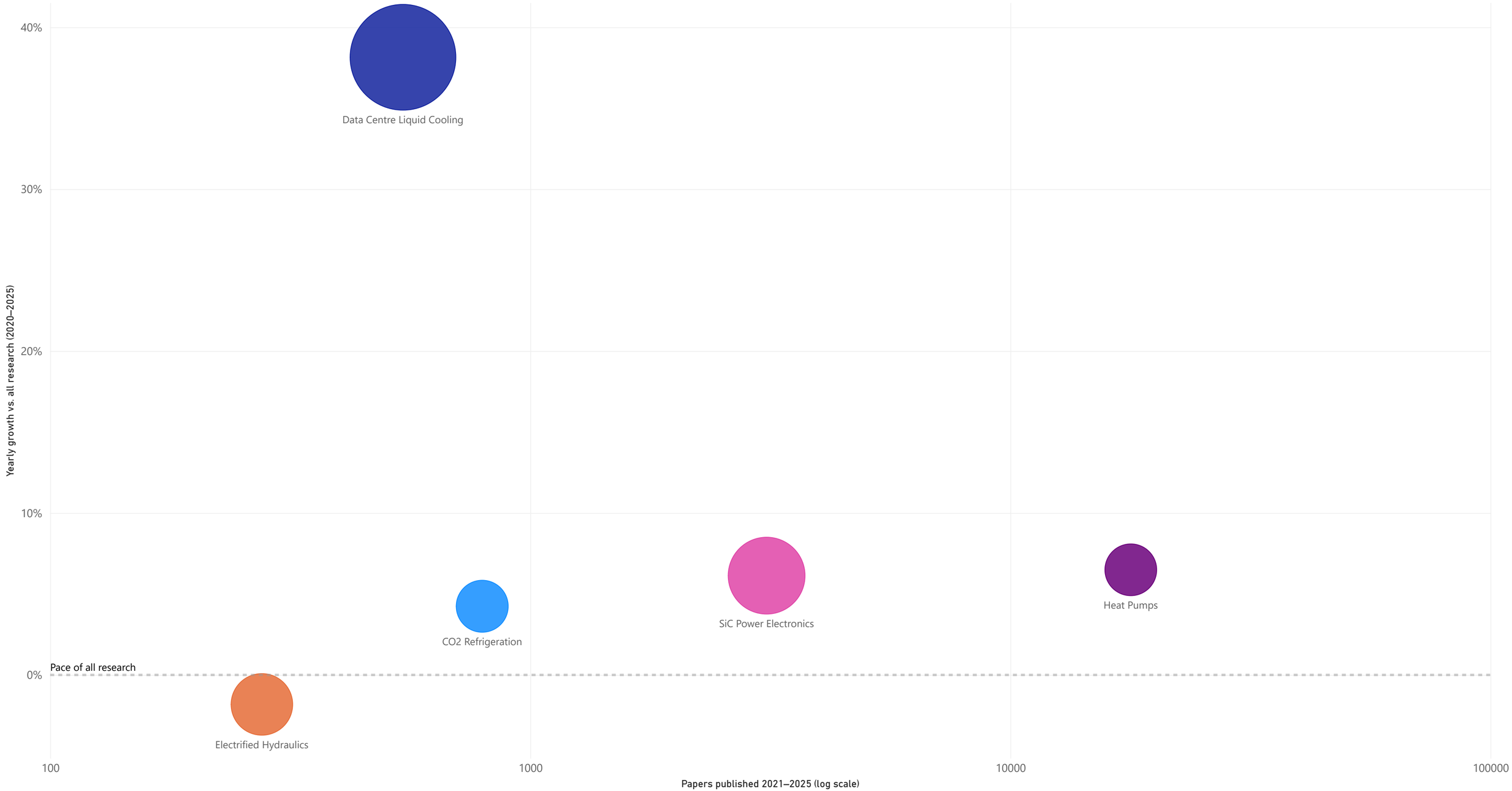
Data centre liquid cooling is the fastest-rising technology, growing 38% a year faster than research overall

Bubble size = share of research from companies (a sign of closeness to market)



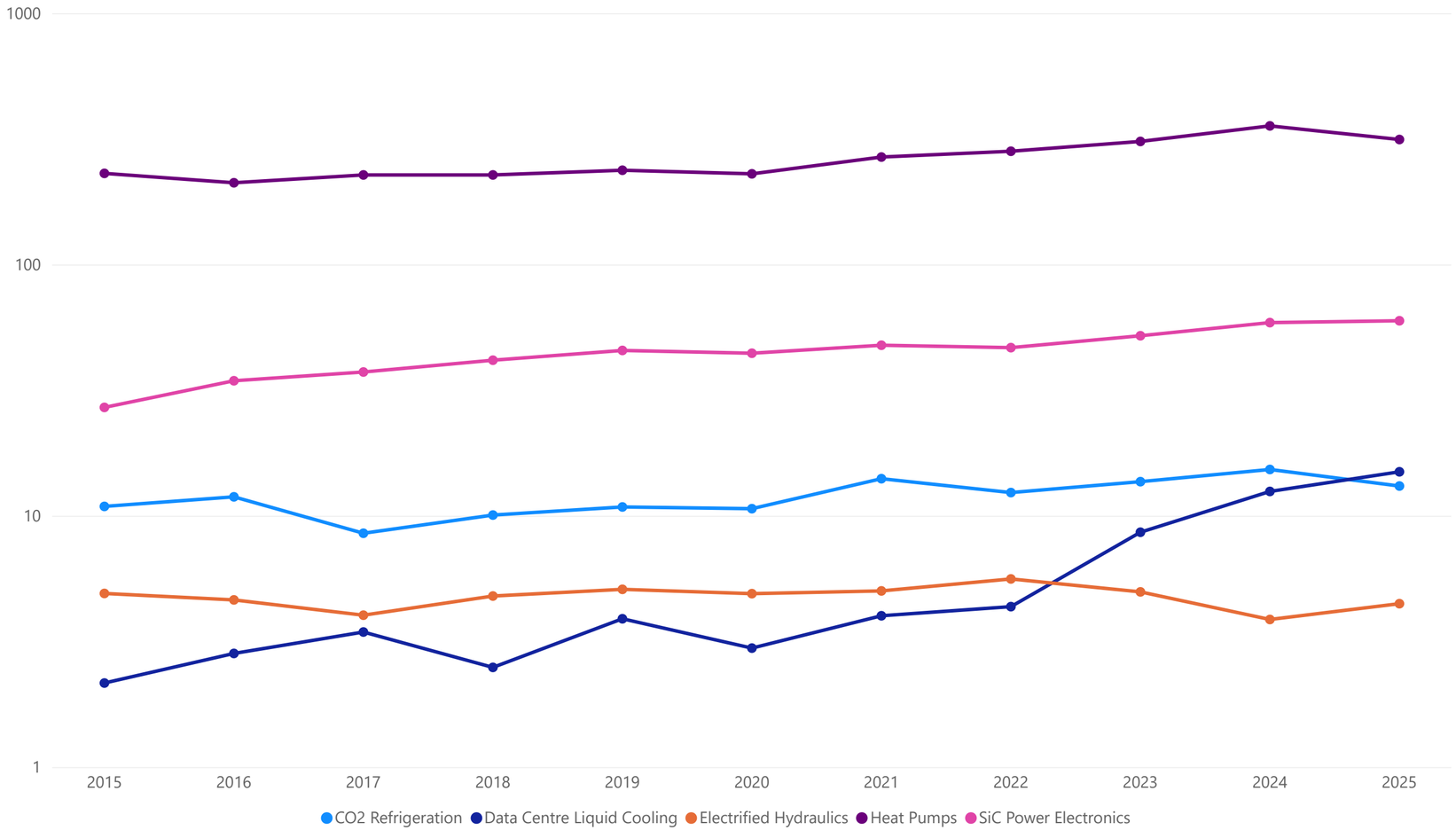
CO2 Refrigeration | **Data Centre Liquid Cooling** | Electrified Hydraulics | Heat Pumps | SiC Power Electronics

Papers 2021-2025
543

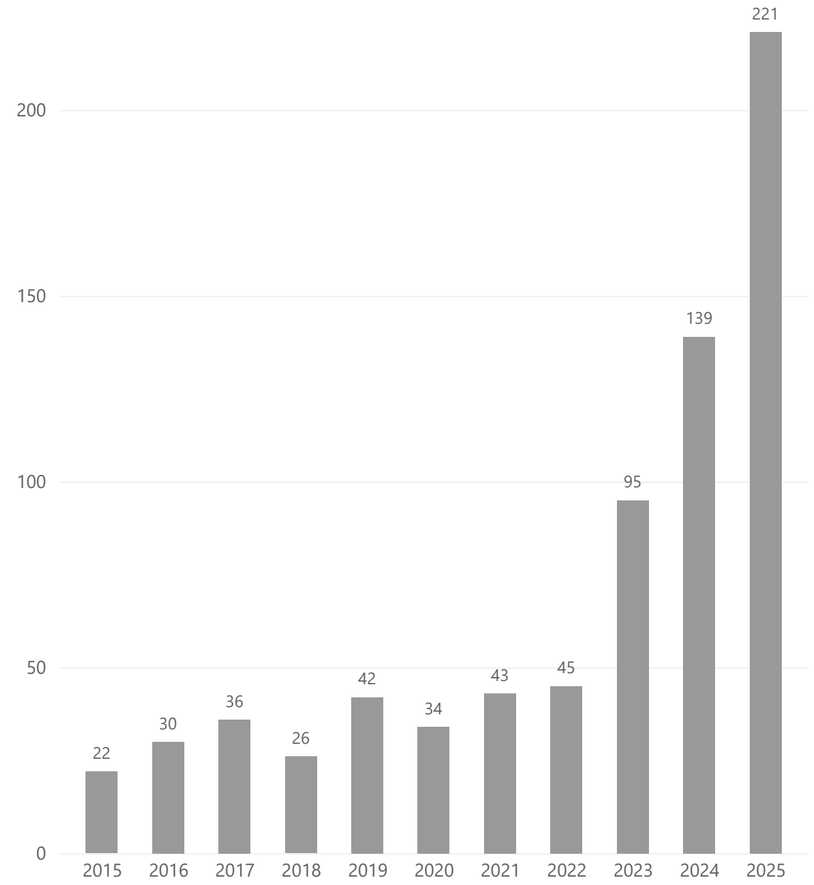
Yearly growth
45.41%

Growth vs. all research
38.14%

Data centre cooling rose from last place to overtake CO2 refrigeration in 2025, growing its share of research 5× since 2020
Papers per million published worldwide (log scale: steeper line = faster growth)



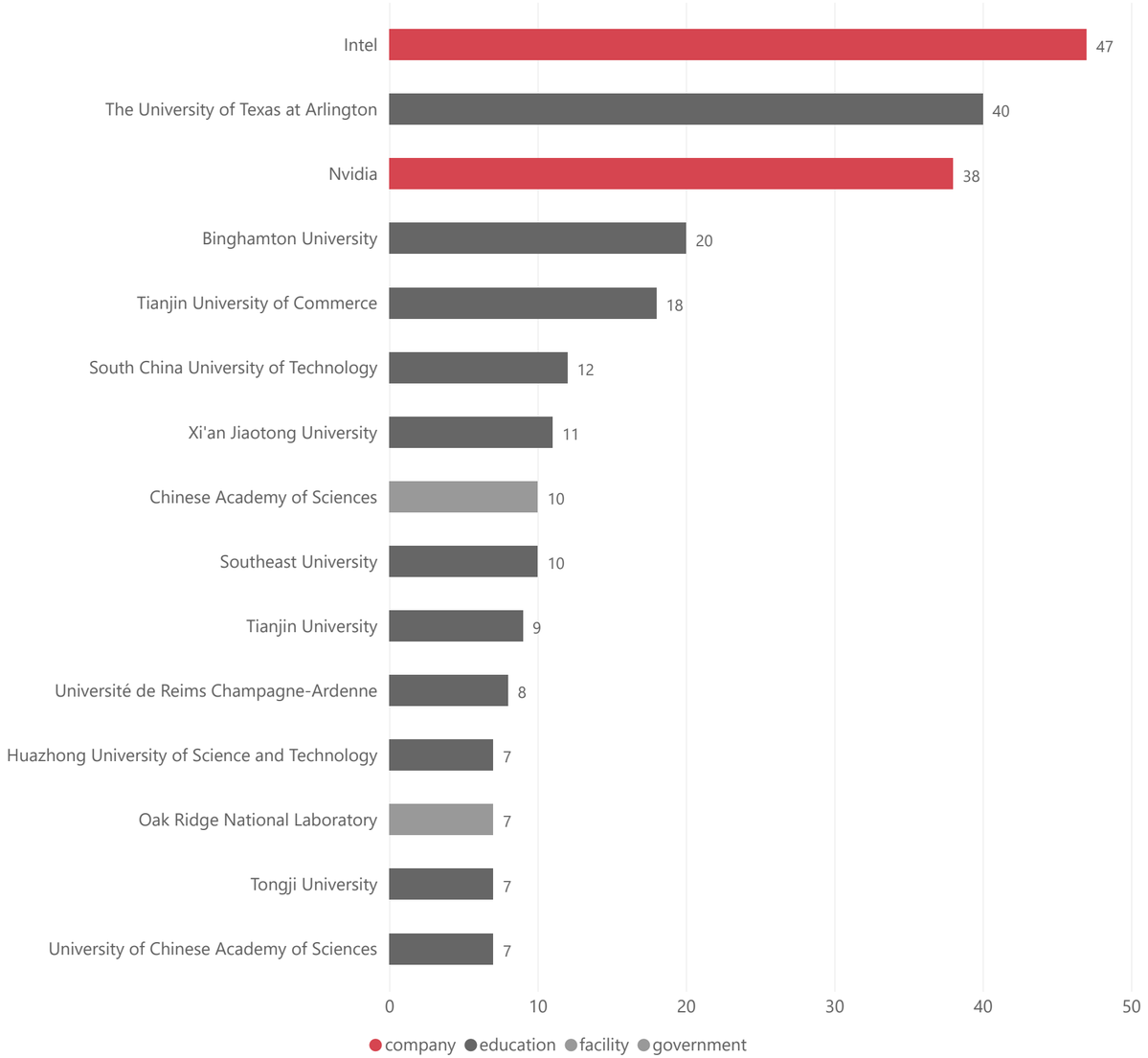
Papers per year (selected technology)
2025 is partly inflated by OpenAlex adding new sources



CO2 Refrigeration | **Data Centre Liquid Cooling** | Electrified Hydraulics | Heat Pumps | SiC Power Electronics

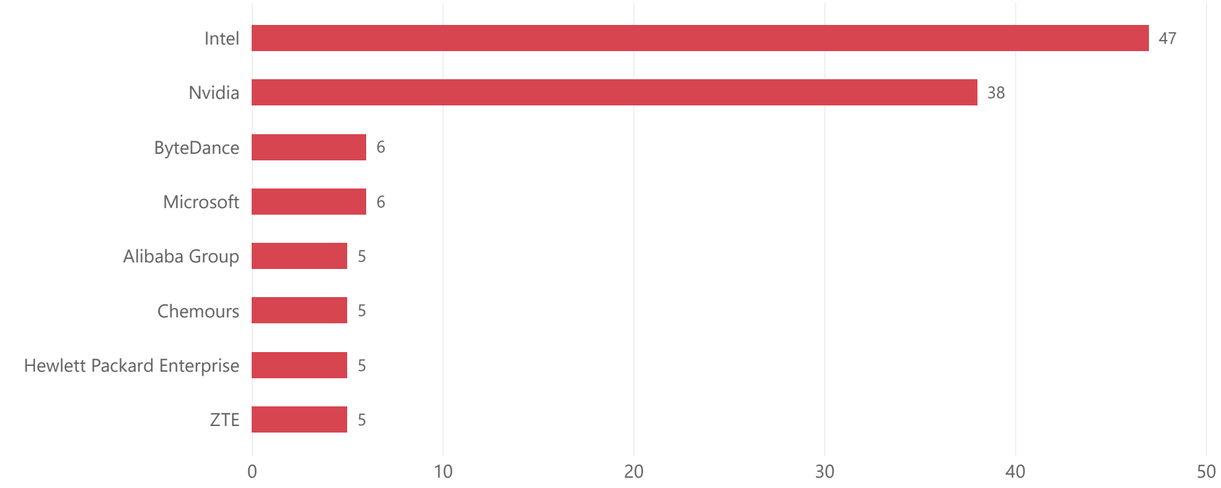
Top 15 organisations in Data Centre Liquid Cooling research (2021–2025)

Papers 2021–2025 · coloured by organisation type, companies highlighted



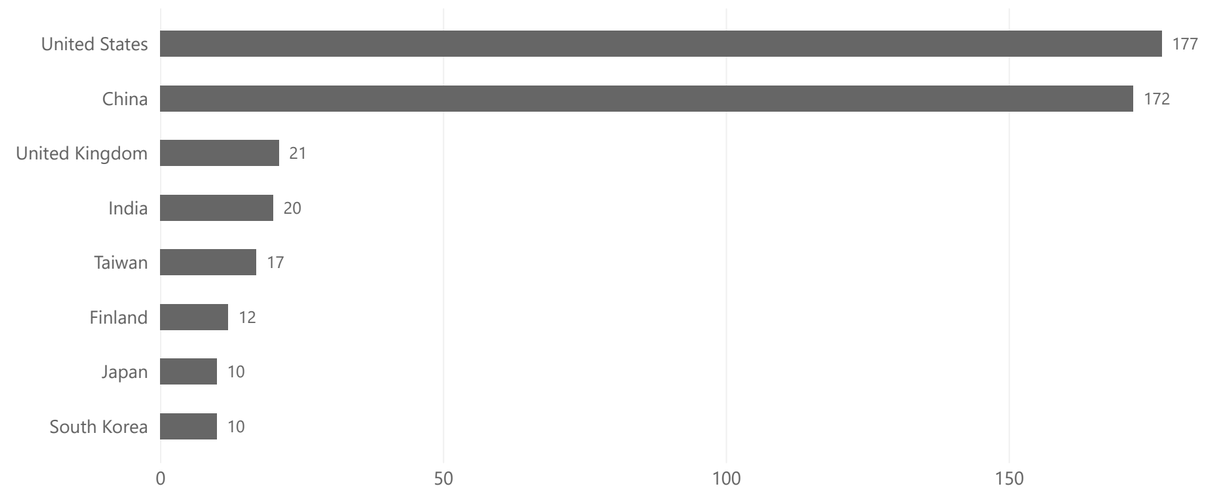
Leading companies in Data Centre Liquid Cooling research (2021–2025)

Companies only · papers 2021–2025



Most active countries

Papers 2021–2025 with at least one author from the country



SiC research is concentrated on lab prototypes, while heat pump research increasingly studies real installations

AI-classified stage of 250 recent papers (2024–2025) · stage of published research, not market maturity

